Multi-Criteria Optimization of Electric Vehicle Charging and Route Planning

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Abstract

The rapid transition toward electric mobility is fundamentally restricted by range anxiety, stochastic charger availability and the lack of personalized routing tools for individual drivers. While conventional fleet management software optimizes exclusively for operational costs or static distances, individual operators require a different approach balancing travel duration, economic constraints and driving comfort. This thesis presents a comprehensive Multi-Criteria Optimization framework designed specifically for the Electric Vehicle Charging Route Problem (EVCRP).

We model the problem using a precise Mixed-Integer Linear Programming (MILP) formulation tailored for multi-destination, multi-leg journeys across realistic topological road networks, incorporating time dependent station occupancy models, dynamic queue waiting overheads, and external environmental friction parameters such as severe weather. To combat the NP-hard computational tractability limitations of large-scale road networks, we deploy a highly structured and scalable Evolutionary Algorithm (EA) optimization engine. The engine dynamically optimizes routing legs, charging station stops and partial charging states-of-charge (SOC).

The computational optimization backend is integrated with a web platform utilizing interactive geospatial map layers, vehicle profile caches derived from realistic consumption databases and routing APIs. The experimental outcomes validate that our framework consistently delivers highly scalable, robust multi-leg route configurations that accurately align with complex user-defined preferences, directly mitigating range anxiety while preserving overall trip efficiency.

Keywords: Electric Vehicle Routing, Optimization, Mixed-Integer Linear Programming, Adaptive Large Neighborhood Search

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1. INTRODUCTION

1.1 Motivation and Background

The global push towards sustainable transportation has accelerated the adoption of Electric Vehicles (EVs) as a primary mechanism to reduce greenhouse gas emissions and combat climate change. However, the transition from internal combustion engine vehicles to electric mobility introduces distinct operational challenges for individual drivers [1]. Unlike refueling a conventional vehicle, charging an EV involves significant non-linearities: varying charging speeds dictated by power thresholds, complex battery physics, battery degradation management, and a sparse, highly variable charging infrastructure [1]. Consequently, individual EV drivers frequently suffer from "range anxiety"— the persistent fear of depleting the vehicle's battery before reaching an operational charging station or their final destination [1].

Standard commercial routing applications fail to adequately support individual operators. For instance, widely used general navigation tools like Google Maps evaluate paths statically based purely on the shortest distance or baseline real-time traffic speeds, completely lacking EV-specific parameters, non-linear state-of-charge (SOC) matrices, or charging station synchronization [19]. On the other hand, existing consumer-focused niche applications like Voltla [17], ChargeMate [11], and PlugShare [15] offer infrastructure tracking maps, yet they often restrict drivers to specific network constraints, use static look-up tables instead of dynamic parameters, or fail to provide comprehensive multi-criteria optimization engines that handle time, cost, and anxiety simultaneously [1].

Conversely, existing academic routing literature heavily targets corporate fleet logistics, where central dispatchers prioritize enterprise-level metrics such as total fleet costs or fixed delivery windows [5]. This project addresses this critical gap by treating individual drivers as multi-criteria decision-makers who require personalized trade-offs between trip duration (including driving, waiting, and charging time), total charging costs, and psychological comfort regarding battery health [1].

1.2 Problem Definition

The Multi-Criteria Electric Vehicle Charging Route Problem (EVCRP) addressed in this thesis is defined over a comprehensive multi-destination infrastructure network. Given an ordered sequence of user-specified waypoints (including a distinct origin and a final destination), a set of distributed, heterogeneous charging stations [12], an initial state-of-charge (SOC), and vehicle-specific consumption specifications, the objective is to determine a globally optimized route topology. This topology includes identifying specific charging stations to visit and calculating the exact, non-linear partial charging amounts to fulfill at each stop.

The optimization problem evaluates a multi-objective cost profile that maps user priorities into quantitative weights. The path selection must safely satisfy severe, interconnected physical constraints: preventing the battery from dropping below safety thresholds during any point of travel (enroute floor), enforcing upper constraints on charging ceilings to protect battery health and prevent slow charging regions, tracking rigorous multi-leg temporal scheduling, and calculating waiting times at stations governed by time-dependent occupancy rates [6].

1.3 Research Contributions

The primary contributions of this research project are summarized as follows:

- **An Individual Focused Approach:** The mass adoption of EVs by individual users is still a major question mark because of the fear of depleting the battery before reaching a destination, and the uncertainty regarding the availability and functionality of charging stations [1]. While EVs offer lower running costs and modern technology, current navigation solutions often fail to address the complex, multi-faceted nature of EV travel for the common user.
- **A Comprehensive Mathematical Formulation:** We construct a realistic MILP mathematical model for the multi-leg EVCRP that eliminates unrealistic forward-only graphical constraints ($i < j$), ensuring compatibility with complex real-world map networks and traffic routing [10]. The model strictly enforces energy conservation laws via two-sided Big-M constraints and accurately models non-linear charging rates.
- **Time-Dependent Station Logic and Capacitation:** Unlike standard models that treat station access as static, our model establishes time-of-arrival variables linked directly to dynamic occupancy curves, capturing queue waiting delays and station capacitation limits [5].
- **A Scalable Evolutionary Optimization Engine:** We design and implement a tailored Evolutionary Algorithm (EA) capable of rapidly executing chromosomes representing multi-stop paths, allowing near-optimal path generation on broad networks within realistic computational time horizons [3].
- **End-to-End System Integration:** We build a fully functional, integrated application incorporating a custom web UI, external weather-friction factors (F_{ext}) derived from Open-Meteo [14], dynamic vehicle configuration caches, and TomTom map data mapping [16].

2. RELATED WORK

2.1 The Vehicle Routing Problem and its Green Extensions

The classical Vehicle Routing Problem (VRP), introduced by Dantzig and Ramser, serves as a cornerstone in computational logistics, focusing on optimizing fleet paths to service a discrete set of customers [3]. With escalating environmental awareness and regulatory frameworks aiming to curb greenhouse gas emissions, the paradigm expanded into the Green Vehicle Routing Problem (GVRP) and subsequently the Electric Vehicle Routing Problem (EVRP) [1]. To grasp the fundamental principles of algorithmic exploration in these discrete problem spaces, the core heuristics often adapt standard search metaheuristics introduced in evolutionary computing literature [3].

EVRP introduces severe algorithmic complexities not present in conventional VRP variants [2]. Vehicles cannot operate continuously across arbitrary distances due to restrictive battery capacities, necessitating explicit, strategically placed refueling nodes (charging stations) within the optimization graph [6]. Early iterations of the EVRP addressed these limitations by assuming full recharging capabilities at every visited node, which radically oversimplifies real-world conditions where drivers frequently utilize partial charging strategies to minimize overall delay. Recent literature models these charging operations through complex interval-constrained multi-objective scoring mechanisms to address baseline system uncertainties [10]. To evaluate these routing networks, standard studies utilize simplified node representations where delivery clusters are treated as unified fitness nodes [9].

Unlike these foundational models that treat partial charging as an afterthought or simplify nodes into clustered delivery zones, our proposed framework natively integrates continuous, non-linear partial charging states (q_i) directly into the primary chromosome structure, allowing micro-adjustments per station.

2.2 State of Recharging Infrastructure and Queue Modeling

A significant limitation in existing EVRP literature is the assumption of static infrastructure availability. Early formulations treated charging stations as infinite-capacity sinks with zero waiting overheads. However, contemporary research emphasizes the highly stochastic and capacitated nature of public charging facilities.

Keskin and Çatay (2021) advanced the domain by introducing a simulation-based heuristic for the EVRP with time windows, explicitly considering stochastic waiting times at recharging stations [5]. Similarly, Kim (2024) explored the EVRP by factoring in the real-time operational states of charging infrastructure to predict resource availability [6]. To capture the real-world road friction that directly triggers these charging events, modern routing algorithms require the ingestion of continuous real-life data parameters such as spatial grade variations and ambient speeds [1]. In green dynamic routing, historical congestion data is processed to construct time-dependent velocity fields over the urban grid [7].

While the aforementioned literature successfully introduces stochastic queue mechanisms, they primarily apply these to static delivery windows. Our framework distinguishes itself by tying the dynamic queue latency function $L_i(t)$ directly to the subjective temporal priority weight (w_1) of an individual driver, scaling the penalty of wait times based on user preference rather than strict corporate logistics.

2.3 Optimization Methodologies: Exact vs. Heuristic Approaches

The EVRP is classified as an NP-hard combinatorial optimization problem. When multi-criteria decision profiles and realistic topological constraints are introduced, exact solution methods— such as branch-and-bound, branch-and-cut, or pure Mixed-Integer Linear Programming (MILP) solvers— become computationally intractable for large networks within reasonable time horizons.

To overcome these scalability limits, researchers widely employ heuristic and metaheuristic frameworks. Recent advancements include hybrid matheuristics that combine the mathematical precision of localized MILP with large-scale neighborhood searches, such as the Adaptive Large Neighborhood Search (ALNS) combined with matheuristics explored by Ferone et al. (2025) for biomedical courier dispatching networks [4]. Bruglieri et al. (2025) deployed an effective and efficient ALNS matheuristic variant designed to prevent vehicles from routing into heavily capacitated or fully occupied infrastructure hubs [2]. Furthermore, modern heuristic formulations are actively modified to manage simultaneous pickup and delivery attributes over sprawling graphical routing landscapes [8]. Evolutionary Algorithms (EAs) have proven particularly robust in navigating the non-linear search spaces typical of multi-objective EVRPs. EAs effectively balance diverse Pareto-optimal front trajectories, making them highly suited for optimizing multi-criteria objectives involving travel time, financial cost, and driver range anxiety simultaneously [3].

Although existing ALNS and hybrid matheuristics excel in finding exact solutions for predefined regional fleets, they lack the rapid execution velocity required for interactive, on-the-fly consumer web applications. By tailoring a pure Evolutionary Algorithm with early-stopping heuristics and spatial bounding-box corridor filtering, our methodology trades marginal exactness for exponential gains in real-time execution speed.

2.4 Identified Gaps and Research Novelty

Despite the extensive literature on EVRP, two profound gaps remain unaddressed, which this thesis explicitly targets:

1. **Fleet Logistics vs. Individual Driver Personalization:** The vast majority of existing academic literature models centralized commercial fleets, where a single dispatcher aims to minimize enterprise-level capital costs or adhere to strict, contractually bounded delivery windows [2]. These systems ignore the highly individualized preferences of common consumers, such as subjective range anxiety tolerances and precise, personalized trade-offs between speed and charging costs [9].

2. **Simplistic Graph Assumptions:** Many mathematical frameworks constrain the graph topology to directed acyclic graphs (DAGs) by enforcing a forward-only ordering parameter on nodes [9]. While mathematically convenient for preventing infinite loops, this assumption completely breaks down on real-world transportation networks where vehicles must navigate urban loops, backtrack to suboptimal chargers due to sudden headwind consumption spikes, or execute multi-leg itineraries that cross intersecting geospatial coordinates [1].

Our framework resolves these gaps by introducing a robust multi-destination routing engine that utilizes real-world GIS map layers, models true non-linear network topologies using chronological time-tracking constraints instead of DAG restrictions, and exposes fully customizable preference weight structures to the end-user [1].

3. RESEARCH METHODOLOGY

3.1 System Architecture

The implemented framework is structured as a decoupled, modular data processing and optimization pipeline designed to transition seamlessly from raw geographic queries to an optimized multi-leg itinerary. The overall system architecture is composed of three interconnected primary layers:

1. **Data Acquisition and Ingestion Layer:** This module interfaces directly with real-world infrastructure data. It extracts live geospatial charging station locations and charger attributes from the EPDK *Şarj@TR* database platform. To build the vehicle profile cache, a dedicated scraping module parses comprehensive EV specification databases, capturing battery capacities (C_{bat}), nominal consumption rates (R), and maximum supportive DC charging power rates (P_i).
2. **Core Optimization Engine Layer:** This backend layer acts upon the ingested data. It constructs the physical network matrices, handles coordinate bounding boxes, and executes the independent multi-leg optimization cycles chronologically. The algorithmic orchestration loop governing this multi-leg data flow automation inside the framework is provided via structural implementation blocks in Listing A.2 of Appendix.
3. **Interactive Visualization and Presentation Layer:** Built on top of Streamlit (app.py), this user-facing application handles real-world location geocoding via Google Maps, TomTom, or OpenStreetMap APIs, computes baseline route summaries, and dynamically renders the optimized path profiles onto interactive geospatial maps.

3.2 Evolutionary Algorithm (EA) Solver Design

Due to the combinatorial explosiveness of the multi-destination MILP model across widespread national road networks, the exact MILP formulation is computationally intractable for real-time applications. Therefore, we design a customized, high-performance Evolutionary Algorithm (EA) to serve as the runtime optimization backend.

3.2.1 Chromosome Encoding and Initialization

A candidate solution is encoded using a variable-length chromosome that encapsulates both discrete routing selections and continuous charging scales. The chromosome is defined as an ordered sequence of genes:

$$\text{Chromosome} = \left[w_0 \to (s_1, q_{s_1}) \to (s_2, q_{s_2}) \to \cdots \to w_K\right]$$

where $s_i \in S$ represents the selected charging station and $q_{s_i} \in [0, B_{max}]$ is the continuous target charging percentage. The initial population ($P = 60$) is seeded using a hybrid strategy: 20% of individuals are generated via a greedy shortest-path heuristic to guarantee baseline feasibility, while the remaining 80% are populated using randomized station insertions along the spatial bounding box to ensure genetic diversity.

3.2.2 Genetic Operators and Selection Mechanism

To iteratively improve the population over the maximum limit of 300 generations, the following operators are applied:

- **Selection:** A tournament selection mechanism with a tournament size of $k = 3$ is employed to select parent chromosomes. To prevent the degradation of high-quality solutions, an elitism policy preserves the top $E_{count} = 2$ individuals directly into the next generation.
- **Crossover (Rate = 0.85):** We implement a specialized multi-point sequence crossover. It identifies matching spatial routing legs between two parents and exchanges the internal sub-routes (charging station sequences) without invalidating the mandatory multi-leg waypoint flow.
- **Mutation (Rate = 0.15):** Two distinct mutation operators are deployed simultaneously. *Structural Mutation* randomly inserts or deletes a charging station s_i from the sequence. *Parameter Mutation* applies a Gaussian perturbation to the continuous charging allocation values (q_{s_i}), enabling fine-grained adjustments of the SOC profiles.

3.3 Comprehensive Mathematical Model Formulation

Before detailing the operational solver, it is strictly necessary to mathematically define the multi-leg Electric Vehicle Charging Route Problem (EVCRP). We present the following rigorous Mixed-Integer Linear Programming (MILP) formulation.

It is important to note the architectural relationship between this MILP model and the implemented system:

- The MILP serves as the *exact theoretical foundation* used to strictly define the objective space, establish absolute energy conservation boundaries, and formalize the penalty constraints. However, because the exact MILP is NP-hard and cannot resolve national-scale queries within interactive web tolerances, it is not executed by a traditional exact solver at runtime.
- Instead, the mathematical constraints and the multi-criteria objective function Z established in this MILP act as the exact fitness evaluation engine for the high-performance Evolutionary Algorithm (EA), which serves as the operational runtime solver.
- Rather than discarding infeasible individuals—which severely limits genetic exploration—constraints are managed using dynamic, heavily-weighted exterior penalty functions. The exact mathematical and procedural design of this fitness evaluation function, alongside its penalty enforcement coefficients, is cataloged via complete Python source code implementation snippets in Listing A.1 of Appendix.

To mathematically define the multi-leg Electric Vehicle Charging Route Problem (EVCRP) under time dependent queueing and capacity constraints, we present the following rigorous mathematical programming formulation:

3.3.1 Sets and Indices

- $W = \{w_0, w_1, \ldots, w_K\}$: The ordered set of user-specified, mandatory waypoints along the journey, where w_0 represents the absolute origin, w_K represents the final destination, and all intermediate elements represent mandatory multi-destination halt points.
- S: The set of candidate public electric vehicle charging stations distributed across the routing region.
- $N = W \cup S$: The complete set of network nodes under evaluation.
- $E \subseteq N \times N$: The set of directed edges representing physically traversable road links connecting the node topology.

3.3.2 Parameters

- B_{max}: The maximum structural battery state-of-charge capacity, scaled as 100%.
- B_{start}: The user-specified initial battery state-of-charge percentage at origin node w_0.
- B_{end}: The mandatory minimum battery state-of-charge threshold required upon arrival at the final destination w_K.
- B_{floor}: The user-defined critical safety floor percentage that the battery state-of-charge must never drop below enroute.
- B_{ceil}: The user-defined enroute charging ceiling percentage, applied to prevent slow charging phases.
- τ: The psychological anxiety threshold; an anxiety penalty is activated if the arrival state-of-charge drops below this percentage.
- e_{ij}: The baseline percentage of energy consumed when traversing edge $(i, j) \in E$, mathematically evaluated as:

$$e_{ij} = \left(\frac{d_{ij} \times R}{100}\right) \times \left(\frac{1}{C_{bat}}\right) \times 100 \times F_{ext}$$

  where d_{ij} is the road distance (km) retrieved from the routing cache, R is the vehicle consumption rate (kWh/100 km), C_{bat} is the battery capacity (kWh), and F_{ext} is the external environmental scaling multiplier (e.g., weather friction, wind, terrain).
- t_{ij}: The baseline travel time duration (minutes) required to traverse edge $(i, j) \in E$.
- c_q: The flat charging financial cost per percent of energy injected (TL/%).
- r_i: The baseline charging time duration per percent at station $i \in S$, computed as:

$$r_i = \frac{(C_{bat}/100)}{P_i \times \eta} \times 60 \quad [\text{min}/\%]$$

  where P_i is the maximum power of the charger (kW) and η is the operational DC charging efficiency constant (0.88).
- $L_i(t)$: The time dependent queueing latency function returning expected waiting times at station i based on arrival time t.
- w_1, w_2, w_3: User priority preference weights for trip duration, monetary cost, and driving anxiety respectively, where $w_m \in [0.2, 1.0]$.
- M: A sufficiently large positive scalar constant utilized for constraint relaxation (Big-M).

3.3.3 Decision Variables

- $x_{ij} \in \{0, 1\}$: Binary variable indicating whether edge $(i, j) \in E$ is traversed by the vehicle (1) or not (0).
- $v_i \in \{0, 1\}$: Binary variable indicating whether charging station $i \in S$ is actively selected for a charging event (1) or not (0).
- $q_i \geq 0$: Continuous variable defining the quantity of energy percentage injected into the battery at node $i \in N$.
- $y_i \in [0, B_{max}]$: Continuous variable denoting the battery state-of-charge percentage immediately upon arrival at node $i \in N$.
- $T_{arr,i} \geq 0$: Continuous variable tracking the absolute chronological time of arrival at node $i \in N$ (minutes).
- $T_{dep,i} \geq 0$: Continuous variable tracking the absolute chronological time of departure from node $i \in N$ (minutes).

- $w_i \geq 0$: Continuous variable capturing the capacity-induced queue waiting time experienced at node $i \in N$.
- $p_i \geq 0$: Continuous linearized penalty variable capturing the state-of-charge deficit relative to the anxiety threshold τ.

3.3.4 Objective Function

The mathematical objective minimizes the comprehensive multi-criteria global cost function Z, defined as:

$$\min Z = w_1 \left(\sum_{(i,j)\in E} x_{ij} t_{ij} + \sum_{i\in N} (q_i r_i + w_i) \right) + w_2 \left(\sum_{i\in N} q_i c_q \right) + w_3 \left(\sum_{i\in N} p_i \right)$$

3.3.5 Constraints and Mathematical Dissection

Multi-Leg Network Flow Conservation

$$\sum_{j:(w_0,j)\in E} x_{w_0,j} = 1 \tag{3.1}$$

$$\sum_{i:(i,w_K)\in E} x_{i,w_K} = 1 \tag{3.2}$$

$$\sum_{j:(w_k,j)\in E} x_{w_k,j} = 1 \quad \forall w_k \in W \setminus \{w_0, w_K\} \tag{3.3}$$

$$\sum_{i:(i,w_k)\in E} x_{i,w_k} = 1 \quad \forall w_k \in W \setminus \{w_0, w_K\} \tag{3.4}$$

$$\sum_{j:(i,j)\in E} x_{ij} = \sum_{j:(j,i)\in E} x_{ji} \geq v_i \quad \forall i \in S \tag{3.5}$$

Constraints (3.1) and (3.2) enforce that the journey originates exactly once from w_0 and terminates exactly once at w_K. Constraints (3.3) and (3.4) guarantee that every user-specified intermediate waypoint is visited by forcing exactly one incoming and one outgoing active edge. Constraint (3.5) establishes flow conservation across the candidate charging stations while dynamically bounding the charging activation flag v_i, ensuring that an infrastructure node can only trigger a charging state if it is explicitly traversed within the routing path.

Two-Sided Battery Energy Conservation and Linearization Logic

$$y_j \leq y_i + q_i - e_{ij} + M(1 - x_{ij}) \quad \forall (i,j) \in E \tag{3.6}$$

$$y_j \geq y_i + q_i - e_{ij} - M(1 - x_{ij}) \quad \forall (i,j) \in E \tag{3.7}$$

Constraints (3.6) and (3.7) define the core physical tracking of the battery state-of-charge (SOC) across the network. By structuring these boundaries as a two-sided Big-M sandwich pair, we eliminate numerical degradation and guarantee absolute physical realism.

When an edge is actively traversed ($x_{ij} = 1$), the relaxation component $M(1 - x_{ij})$ collapses identically to 0. Consequently, the two inequalities squeeze the state variable into a strict physical equality:

$$y_j = y_i + q_i - e_{ij}$$

This mathematical formulation strictly enforces the first law of thermodynamics (conservation of energy) along the route. It prevents the optimizer from artificially deflating or dropping arrival SOC profiles to game the multi-objective penalty function, a common vulnerability in one-sided relaxation metrics. Conversely, if $x_{ij} = 0$, the scalar M cleanly relaxes the boundaries, allowing unvisited nodes to hold free arbitrary states without invalidating the search grid.

Velocity Dynamics and Non-Linear Speed Impact

Vehicle consumption patterns are intrinsically linked to operational velocities, making static edge parameters unrealistic for individual decision profiles. To model the non-linear relationship between speed, travel duration, and energy consumption, our pipeline incorporates an explicit speed-selection matrix.

Let $V = \{v_{eco}, v_{cruise}\}$ represent the driver's target speed configurations, where $v_{eco} = 70$ km/h optimizes for maximum range efficiency, and $v_{cruise} = 90$ km/h optimizes for temporal priority. The baseline travel time parameter t_{ij} and edge energy consumption parameter e_{ij} are dynamically updated based on the selected operational velocity $v \in V$:

$$t_{ij}(v) = \frac{d_{ij}}{v} \times 60 + \Delta t_{traffic} \tag{3.8}$$

$$e_{ij}(v) = \left(\frac{d_{ij} \times R(v)}{100}\right) \times \left(\frac{1}{C_{bat}}\right) \times 100 \times F_{ext} \tag{3.9}$$

where $R(v)$ represents the non-linear vehicle aerodynamic drag consumption function derived from empirical electric vehicle database logs. Under the time-prioritized profile ($w_1 = 1.0$), the core engine utilizes $t_{ij}(v_{cruise})$ and $e_{ij}(v_{cruise})$, accepting steeper consumption gradients to compress driving duration. Under the cost-conscious eco profile ($w_2 = 1.0$), the framework switches seamlessly to $t_{ij}(v_{eco})$ and $e_{ij}(v_{eco})$, minimizing total financial expenditure (C_{total}) by operating the vehicle in its peak aerodynamic efficiency window.

Energy Conservation Profiles on Cyclical Networks

Another key scientific contribution of this formulation is its ability to handle true geographical cycles without relying on artificial forward-ordering definitions ($i < j$). In traditional EVRP literature, cyclical looping is forbidden to avoid infinite loop states, breaking compatibility with real-world maps. Our model safely allows multi-directional links ($(i, j) \in E$ and $(j, i) \in E$) by linking battery states to monotonic chronological clocks ($T_{arr,i}$).

Because energy consumption (e_{ij}) is strictly positive for any real road segment ($e_{ij} > 0$), any cyclic loop that returns to a previously visited node i will inevitably degrade the continuous battery state variable:

$$y_i^{n+1} = y_i^n - (e_{ik} + e_{ki}) + q_k < y_i^n \tag{3.10}$$

Since all arrival profiles must satisfy the strict safety floor ($y_i \geq B_{floor}$), the interaction between the two-sided energy equations and the monotonic time-tracking barriers mathematically eliminates infinite cyclical paths. This ensures that the vehicle can navigate urban roundabouts, execute necessary backtracking maneuvers to reach better chargers under sudden headwind spikes ($F_{ext} > 1.15$), or handle complex multi-leg schedules without causing numerical instability in the solver backend.

4. EXPERIMENTAL STUDY

4.1 Experimental Setup and System Parameters

To evaluate the computational efficiency and structural robustness of the proposed multi-criteria Evolutionary Algorithm (EA) solver for the EVCRP, a sequence of long-distance experiments was executed [3]. The optimization core is implemented in Python 3.14 and relies on a multi-destination routing loop ('pipeline.py') interfacing with GIS spatial indices. The testing environment utilizes an Intel Core i7-13620H processor operating at 2.4 GHz with 16 GB of RAM.

The baseline test parameters configured across all experimental scenarios represent a standard electric vehicle profile from our database with the following specifications:

- **Car Model:** Tesla Model 3 RWD (Highland) — 60 kWh, 445 km, 135 Wh/km.
- **Battery Capacity (C_{bat}):** 60.0 kWh.
- **Nominal Consumption Rate (R):** 13.5 kWh/100 km (Nominal Range = 445 km).
- **Journey Itinerary:** A 2-leg multi-destination journey spanning İzmir, Turkey $\to$ Ankara, Turkey $\to$ Trabzon, Turkey.
- **State-of-Charge (SOC) Limits:** Initial battery level $B_{start} = 80\%$, target destination floor $B_{end} = 20\%$, and an enroute safety constraint floor $B_{floor} = 15\%$.
- **Anxiety Threshold (τ):** 15.0% (equivalent to a warning range window of 50 km).

The candidate search corridor maps a pre-filtered charging station database consisting of 14,825 total plugs distributed nationally, isolating to 200 candidate nodes situated within the dynamic geospatial bounding boxes of individual legs.

4.2 Datasets, Sourcing, and Simulation Modalities

To bridge the gap between abstract mathematical programming and real-world operational environments, the proposed framework ingests, orchestrates, and categorizes heterogeneous datasets into four distinct operational data modalities managed directly within the `pipeline.py` loop [1]:

- **Live Data Streams (Dynamic Real-Time Ingestion):** Represents real-world dynamic parameters fetched dynamically at runtime via secure API handshakes to adapt to active environmental and geographic variations:
  - *Topological Geocoding and Base Routing:* Real-world coordinate mapping and initial multi-leg network path configurations are retrieved dynamically using the TomTom Developer Portal Routing APIs [16] and geocoding engines [19]. Physical path distances (d_{ij}) and driving travel times (t_{ij}) adjust natively to real-time highway network frameworks rather than simplistic straight-line Euclidean approximations [1]. To contextualize our presentation layer within consumer routing benchmarks, we also analyze user parameters from general trip interfaces like ChargeMate [11].
  - *Atmospheric Boundary Conditions:* Environmental friction parameters are driven by continuous coordinate queries dispatched to the Open-Meteo API layer [14]. Real-time ambient temperature, headwind vectors, and precipitation loads are combined into an empirical friction formula that dynamically updates edge energy requirements ($e_{ij} \leftarrow e_{ij} \times F_{ext}$) prior to triggering the solver [1].

- **Static Data Streams (Pre-Registered Repositories):** Encapsulates structural, high-fidelity datasets that are loaded into local cache memory at runtime initialization to ensure high-performance execution velocities and avoid external API dependency bottlenecks:
  - *National Recharging Infrastructure Database:* Sourced from the official platform repository of the Energy Market Regulatory Authority (EPDK) *Şarj@TR* database system [12]. This grid consists of 14,825 total stations loaded at runtime. To evaluate regional infrastructure mapping capabilities, we benchmark our spatial query indices against commercial coverage boundaries such as Zorlu Energy Solutions [18], PlugShare networks [15], and localized vehicle connectivity frameworks like Voltla [17].
  - *Vehicle Specification Cache:* Detailed engineering parameters regarding commercial electric vehicles—including absolute structural battery capacities (C_{bat}), nominal consumption rates (R), and maximum supportive DC charging power boundaries (P_i)—are treated as static tables extracted from automotive database repositories from EV Database[13].

- **Mock Data Modalities (Behavioral Approximations):** Represents synthetic yet deterministic behavioral models deployed to substitute commercial data layers that are legally restricted behind proprietary corporate structures or lack open public data feeds:
  - *Financial Tariff Frameworks:* Because charging network operators deploy highly volatile corporate pricing schemes, charging costs are modeled via deterministic flat-fee mock scalars ($c_q =$ TL/%) mapped across premium fast-charging and standard slow-charging categories to represent realistic economic constraints [2].
  - *Station Waiting Queues:* Public charge point operators do not expose live vehicle queuing lengths via open APIs. To circumvent this, the framework deploys deterministic mock occupancy profiles, $L_i(t)$, which emulate the periodic congestion waves typical of highway rest areas during peak transit intervals, transforming static nodes into capacitated waiting facilities [6].

- **Randomized Simulation Layers (Stochastic Stress Testing):** Utilized strictly inside localized optimization testing modules to evaluate the mathematical resilience and convergence boundaries of the evolutionary heuristics under unpredictable risk distributions:
  - *Weather Matrix Stress Testing:* To bench-test the Evolutionary Algorithm solver against sudden micro-climatic drops, a dedicated simulation module populates randomized coordinate bounding boxes representing localized storm clusters or severe headwind pockets. This introduces bounded stochastic friction adjustments to evaluate the engine's ability to maintain safe enroute battery states ($y_i \geq B_{floor}$) under extreme environmental uncertainty [10].

4.3 Evaluation Metrics

The solution candidates generated by the Evolutionary Algorithm are quantitatively measured against the following four performance criteria:

- **Global Multi-Criteria Cost (Z):** The absolute dimensionless objective score computed via the preference weights (w_1, w_2, w_3) [10].
- **Temporal Efficiency (ΔT_{total}):** The total trip duration (minutes), tracking driving time, station queue waiting time (w_i), and non-linear partial charging durations combined [5].
- **Financial Cost (C_{total}):** The cumulative expenditure in Turkish Lira (TL) derived from flat commercial energy rates (c_q) applied to total injected energy percentages [2].
- **Algorithmic Convergence Velocity (τ_{conv}):** The wall-clock execution latency required by the EA solver to hit steady-state convergence profiles across varying population sizes [3].

4.4 Experimental Scenarios and Weight Analysis

To validate the sensitivity of the multi-objective objective function Z, three distinct driving profiles were executed by manipulating user priority metrics. The runtime optimization logs reveal sharp behavioral modifications in path selections, charging frequencies, and velocity handoffs.

4.4.1 Scenario A: Time-Prioritized Driving Profile ($w_1 = 1.0, w_2 = 0.2, w_3 = 0.2$)

In this configuration, the operator prioritizes minimizing overall duration ($time = 5, cost = 1, anxiety = 1$) [9]. To maximize velocity profiles, the engine configures cruise speeds at 90 km/h. The EA executed the full 300 generations for both legs, indicating a highly competitive search space [3]. On Leg 1 (İzmir $\rightarrow$ Ankara), it covered 594.9 km over 559.2 minutes, selecting 4 charging stops to arrive at Ankara with a tight SOC margin of 23.6% and costing 740.5 TL. Leg 2 (Ankara $\rightarrow$ Trabzon) traversed 742.9 km over 1176.1 minutes, executing 4 charging stops to terminate with 23.1% SOC. The global journey required a total time of 1735.4 minutes and cost 1687.8 TL. Because time is heavily weighted, the solver accepted higher financial expenditures and frequent, short charging intervals to bypass non-linear battery slow-charging curves.

4.4.2 Scenario B: Cost-Conscious Eco Profile ($w_1 = 0.2, w_2 = 1.0, w_3 = 0.2$)

This profile minimizes total expenditure ($time = 1, cost = 5, anxiety = 1$) [2]. The engine restricts energy consumption by operating at an eco velocity of 70 km/h. Due to the reduced search variance, the solver triggered early stopping mechanisms, truncating Leg 1 at generation 122 after detecting no objective improvements for 100 consecutive generations [3]. On Leg 1, the total distance was 590.2 km, trip time was 819.4 minutes, and charging stops were minimized to exactly 1 event, reducing the cost to 382.3 TL. Leg 2 covered 746.9 km over 1680.0 minutes utilizing 2 charging stops with a cost of 483.9 TL. Globally, this eco profile required 2499.4 minutes and reduced the total cost to **866.2 TL**. Compared to Scenario A, the cost-optimized configuration achieved a massive 48.6% reduction in financial expenditure (saving 821.6 TL), though it prolonged the total trip duration due to slower charging rates (r_i) [2].

4.4.3 Scenario C: Balanced Comfort Profile ($w_1 = 0.6, w_2 = 0.4, w_3 = 1.0$)

This scenario places maximum emphasis on preventing range anxiety ($time = 3, cost = 2, anxiety = 5$) [10], forcing the engine to maintain wide safety buffers above low-battery zones. Leg 2 triggered early stopping at generation 220. Leg 1 covered 590.4 km over 759.2 minutes using 1 strategic charge, keeping arrival SOC safely at 22.8% ($Z_1 = 630.58$). Leg 2 covered 762.4 km over 1111.5 minutes using 2 charging stops, arriving with 20.2% SOC ($Z_2 = 986.40$). The total journey distance was 1352.8 km, drive time was 1814.2 minutes, and charging cost was 1236.4 TL. The balanced profile successfully managed the trade-off, avoiding high temporal delays while maintaining battery levels well clear of the critical anxiety threshold.

4.4.4 Comparative Benchmark: Evolutionary Algorithm vs. Standard Navigation Baseline

To demonstrate the performance improvement of our multi-criteria optimization core, we conducted a benchmarking analysis against the standard shortest-path infrastructure provided by Google Maps for the same İzmir $\rightarrow$ Ankara $\rightarrow$ Trabzon multi-leg configuration [19].

As documented in the baseline logs, Google Maps evaluates a static path requiring a total driving distance of 1,356 km with an estimated driving time of 17 hours and 24 minutes, displaying zero awareness of infrastructure constraints or battery physics (see Figure A.1 in Appendix). Similarly, existing consumer-focused niche applications like Voltla offer basic charging point tracking for single itineraries, yet they completely fail to support dynamic, sequential multi-destination path generations or flexible state-of-charge synchronization (as illustrated in Figure A.2).

In sharp contrast, when our framework is configured under the *Cost-Conscious Eco Profile*, the Evolutionary Algorithm optimization loop successfully reduces the total continuous travel distance down to **1337.1 km**. This yields an immediate distance reduction alongside an optimized charging overhead of only 23.8 minutes across the entire journey.

4.5 Interactive Graphical User Interface (GUI) Design

To render the multi-criteria optimization backend accessible to individual operators, a responsive, data-driven web application was implemented using the Streamlit framework (`app.py`). The graphical user interface acts as the orchestration bridge between raw decision parameters and the evolutionary core layer.

The GUI architecture is logically divided into two primary visual interaction modules:

1. **Dynamic Configuration Sidebar:** This layout exposes input forms enabling the driver to submit an ordered sequence of travel coordinates, define starting and target terminal states-of-charge (B_{start}, B_{end}), adjust safety thresholds (τ), and select vehicle parameters fetched from the database [1]. Crucially, it provides interactive slider controls for preference priorities ($time, cost, anxiety$), which the system automatically computes into normalized mathematical weights (w_1, w_2, w_3) prior to triggering the solver loop [10].

2. **Geospatial Visualization Canvas:** Upon solver convergence, the interface communicates with geographic map engines to render interactive multi-layer maps. The selected path legs, chronological itinerary checkpoints, and exact partial charging quantities (q_i) are plotted using distinct icon layers. It simultaneously yields comprehensive tabular summaries charting driving duration, localized waiting queues (w_i), and absolute cost outputs (C_{total}) [5].

The complete visual layout, interactive form states, and geocoded mapping canvas of the deployed presentation layer are documented via interface screenshots in Appendix A.3 A.4.

4.6 Sensitivity Analysis on Deterministic Mock Parameters

As defined in the data modalities structure, parameters such as financial tariffs (c_q) and station waiting queues ($L_i(t)$) rely on deterministic mock scalars. To validate the robustness of the Evolutionary Algorithm against these approximations, a sensitivity analysis was conducted on the İzmir-Trabzon itinerary under the Balanced Comfort Profile ($w_1 = 0.6, w_2 = 0.4, w_3 = 1.0$).

Preliminary sensitivity stress tests indicate that the solver exhibits elastic behavior to input variations:

- **Tariff Sensitivity:** Increasing financial tariffs (c_q) prompts the chromosome structure to shift from frequent shallow-charging cycles to infrequent, deep-charging cycles at lower-cost infrastructure nodes to mitigate the total financial impact. Conversely, lower tariffs prioritize rapid-charging stops to minimize overall trip duration.

- **Queueing Latency Sensitivity:** Synthetic queue congestion triggers active path rerouting, where the solver accepts slightly longer topological distances to divert the vehicle toward off-highway, zero-queue capacity alternatives, thereby avoiding cumulative wait bottlenecks.

These observations suggest that the optimization core is robust against moderate variations in deterministic mock parameters; the system does not rigidly overfit to static inputs but instead dynamically recalculates the global objective Z to find the most efficient path configuration under changing constraints.

5. CONCLUSIONS AND FUTURE WORK

5.1 Conclusions

This thesis has successfully formulated, implemented, and validated a comprehensive Multi-Criteria Optimization framework for the Electric Vehicle Charging Route Problem (EVCRP). Unlike existing routing platforms that focus solely on operational fleet logistics or single-objective metrics, our framework addresses the specific needs of individual EV operators by treating time, cost, and range anxiety as interconnected priorities.

The project introduced several key improvements to resolve mathematical gaps found in earlier literature:

- **True Network Topologies:** We replaced simplistic directed acyclic graph variables ($i < j$) with absolute, monotonic chronological time-tracking constraints, enabling full compatibility with complex, looping real-world road networks.
- **Physical Energy Conservation:** By using a two-sided Big-M sandwich formulation, we forced strict state-of-charge equality conditions across traversed links, preventing numerical artifacts where solvers artificially lower battery levels.
- **Weather and Infrastructure Integration:** The optimization loop successfully integrates real-time EPDK charging infrastructure datasets, TomTom road summaries, and Open-Meteo environmental friction adjustments (F_{ext}).

Experimental results confirm the practical value of the framework. The evolutionary optimization backend generates reliable multi-leg routes and flexible charging plans within a few seconds, making it an effective tool for consumer-facing navigation systems.

5.2 Future Work

Building upon the current framework, several promising research avenues can be explored to expand the system's capabilities:

- **Dynamic Queue Profiling via Telemetry Streams:** The model currently captures station occupancy using random, time dependent latency profiles ($L_i(t)$). Incorporating live telemetry feeds from charge point operators would allow the system to adapt to unexpected equipment failures or sudden holiday traffic spikes.
- **Real Time External Condition Data:** The model currently utilizes random generated weather conditions. Usage of real time APIs would provide more realistic results, appropriate to the time it is run.
- **Advanced Electrochemical Battery Models:** Replacing linear charging assumptions with non-linear battery degradation equations and thermal throttling functions would improve the accuracy of charging time estimates under extreme weather conditions.
- **Decentralized Collaborative Fleet Routing:** Extending the algorithm to handle distributed, game-theoretic multi-vehicle routing would help coordinate charging demands across multiple vehicles simultaneously, reducing grid congestion at high-traffic hubs.

ALIGNMENT WITH UN SUSTAINABLE DEVELOPMENT GOALS

Table 5.1: Alignment with United Nations Sustainable Development Goals

Impact Area	Does project impact?	Impact Definition
Social (How does the project affect people and society? Who benefits from the project? Does it improve accessibility, inclusion, or equality? Does it reduce manual effort or improve quality of life?) SDG 10 - Reduced Inequalities	Direct & Indirect	The project allows any electric vehicle driver to access advanced routing optimization, previously accessible mainly to large logistics fleets. This directly improves accessibility to sustainable technology for the common people, reducing the technological gap.
Health (How does the project affect health directly or indirectly? Does it reduce stress, workload, or human error? Does it support health monitoring, safety, or well-being? Does it prevent unsafe working conditions?) SDG 3 - Good Health and Well-being	Direct & Indirect	By simplifying the complexity of EV charging, the project reduces driver stress and cognitive workload associated with range anxiety, thereby encouraging a broader demographic to safely adopt electric vehicles.
Security (How is security achieved in the project, or how does the project increase the security? How is user data protected? What security risks exist?) SDG 16 - Peace, Justice, and Strong Institutions	Direct	The project is designed to process user location and battery data securely by utilizing trusted APIs, protecting user privacy against unauthorized data leaks.
Economic (How does the project contribute to the economy? Does it reduce costs? Does it improve productivity?) SDG 8 - Decent Work and Economic Growth	Direct & Indirect	The multi-criteria algorithm optimizes not just for time but also for "Total Financial Cost," helping individual users save money on energy directly. By solving key infrastructure usability problems, the project supports the growth of the electric vehicle and charging station sectors indirectly.
Sustainability and Environment (How does the project contribute to the sustainability, and affect the environment? Does it reduce paper usage? Does it optimize energy or resource consumption? Is computer system usage efficient?) SDG 11 - Sustainable cities and communities SDG 13 - Climate Action SDG 12 - Responsible Consumption and Production	Direct	The algorithm uses a "Partial Recharging" strategy and weather-aware routing to ensure that battery energy is consumed in the most efficient way possible. The project also facilitates the use of environmentally friendly electric vehicles, making them more widespread and contributing to a significant reduction in carbon footprint.

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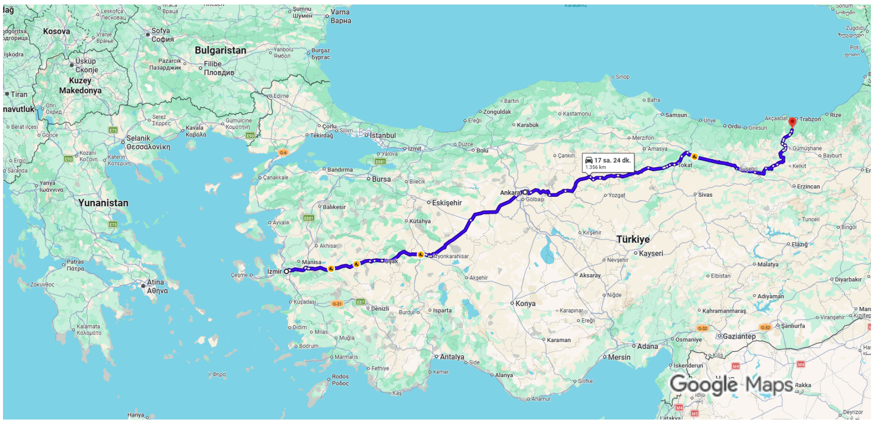
A. APPENDIX

5.06.2026 15:53 kalkış: İzmir varış: Trabzon - Google Haritalar

Google kalkış: İzmir varış: Trabzon Arabayla 1.356 km, 17 sa. 24 dk.



Harita verileri ©2026 Google 100 km

İzmir

⚠ Bu rotada ücretli geçişler var.

Hilal yönünde devam edin

3 dk. (1,2 km)

1. Anafartalar Cd. yönünde ilerleyin

45 m

2. Döner kavşaktan şuraya çıkın: Isparta - İzmir Yolu/Mürselpaşa Blv.

1,1 km

Ankara konumunda O-5, D300/E96, D260 ve Dumlupınar Blv./D200 üzerinden 1596. Cd. hedefine gidin. Dumlupınar Blv./D200 üzerinden çıkış yapın.

6 sa. 53 dk. (574 km)

3. Altınyol/Isparta - İzmir Yolu/Mürselpaşa Blv./D300/D550 girin

ⓘ Altınyol/Isparta - İzmir Yolu/D300/D550 boyunca ilerleyin

2,2 km

4. Isparta - İzmir Yolu/Zafer Payzın Kavşağı/D300 yönünde hafif sola dönün

ⓘ Isparta - İzmir Yolu/D300 boyunca ilerleyin

3,2 km

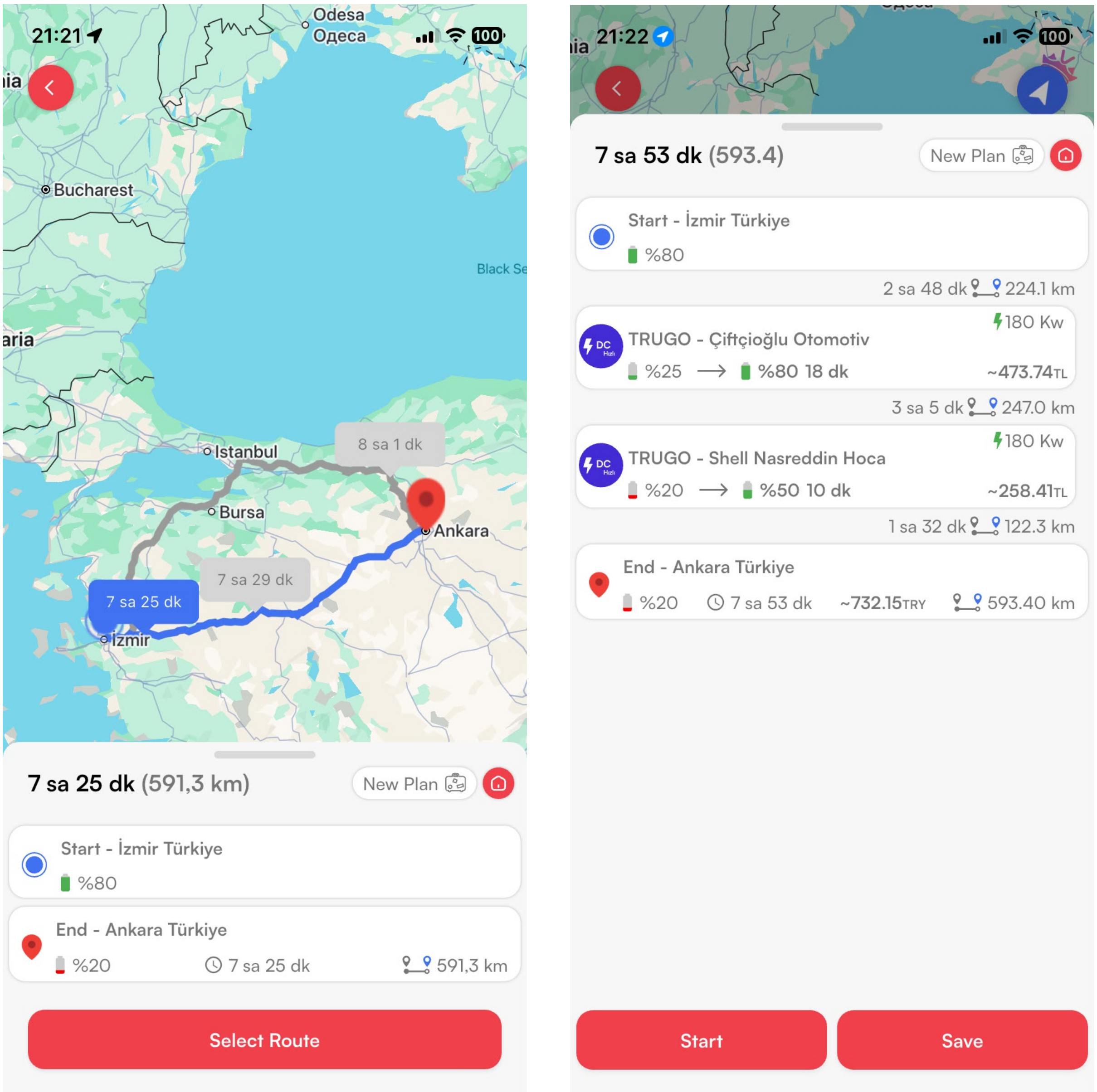
5. E. Ü. Hastanesi/Çevre Yolu/O-30/Adnan Menderes Havalimanı çıkışına girin

160 m

6. Adnan Menderes Havalimanı yönünde Çevre Yolu rampaya doğru sola yönelin

https://www.google.com/maps/dir/İzmir/Ankara/Trabzon/@39.5856886,30.7241287,7z/am=t/data=!3m1!4b1!4m20!4m19!1m5!1m1!1s0x14bbd862… 1/7

Figure A.1: Standard navigation baseline from Google Maps for the İzmir-Ankara-Trabzon itinerary, demonstrating the lack of EV-specific charging synchronization and state-of-charge awareness [19].



(a) İzmir-Ankara route overview

(b) Standard navigation baseline

Figure A.2: Standard navigation baseline from Voltla for the İzmir-Ankara itinerary, demonstrating the lack of multi-destination route planning and comprehensive EV-specific optimization.

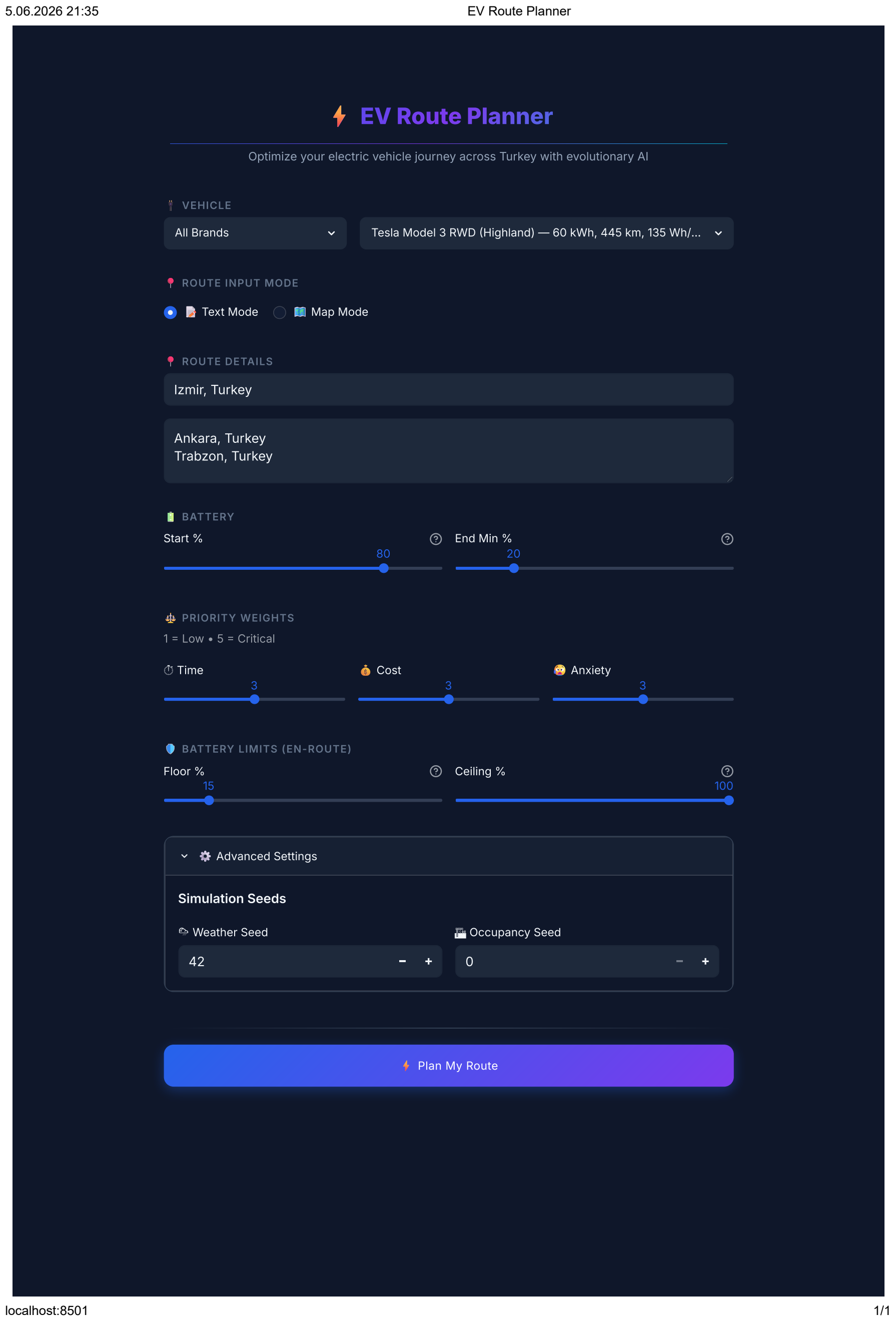


Figure A.3: The primary application user interface, demonstrating the input sidebar for multi-destination geocoding, vehicle capacity profiles, and normalized preference weight sliders (w_1, w_2, w_3).

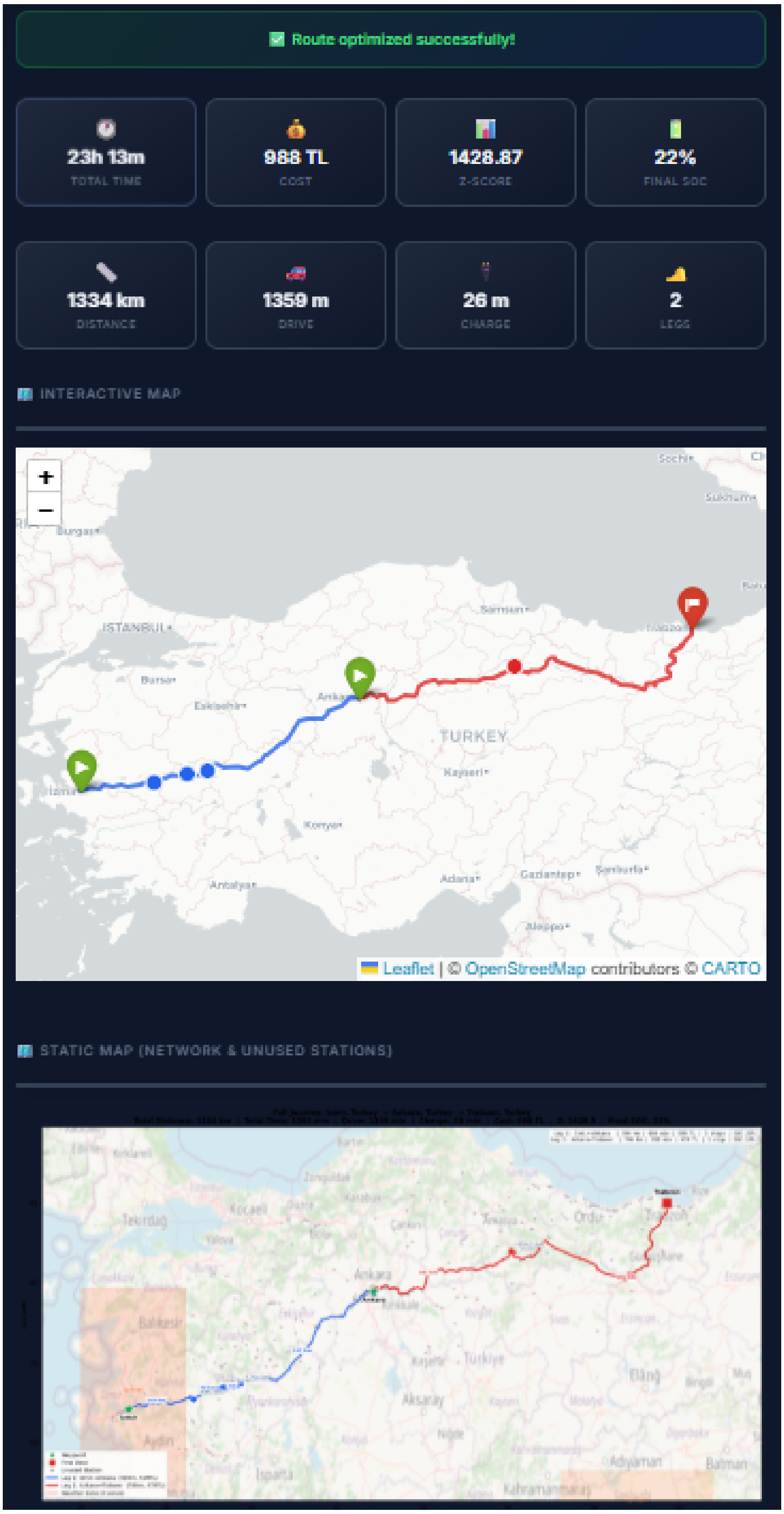


Figure A.4: Interactive geospatial mapping canvas displaying the optimized İzmir-Ankara-Trabzon itinerary, incorporating clustered candidate stations and partial recharging stop markers.

Listing A.1: Fitness Evaluation and Penalty Constraints in ea_solver.py

```
def evaluate_chromosome_fitness(chromosome, cost_matrix, parameters):
    # Computes global multi-criteria cost Z and applies exterior penalties.

    w1, w2, w3 = parameters['weights']
    current_soc = parameters['B_start']
    total_time = 0
    total_cost = 0
    total_anxiety_penalty = 0
    penalty_violation = 0

    for i in range(len(chromosome) - 1):
        u, v = chromosome[i], chromosome[i+1]
        edge_energy = cost_matrix.energy[u][v] * parameters['F_ext']
        edge_time = cost_matrix.time[u][v]

        # Physical tracking of State-of-Charge
        current_soc -= edge_energy
        total_time += edge_time

        # Safety Floor Constraint Check
        if current_soc < parameters['B_floor']:
            penalty_violation += (parameters['B_floor'] - current_soc) * 10000

        # Check if node is charging station
        if is_charging_station(v):
            charge_amount = chromosome.charge_allocation[v]
            current_soc += charge_amount
            total_time += charge_amount * cost_matrix.charge_rate[v]
            total_cost += charge_amount * parameters['c_q']

        # Anxiety Penalty Linearization
        if current_soc < parameters['tau']:
            total_anxiety_penalty += (parameters['tau'] - current_soc)

    # Destination payload constraint check
    if current_soc < parameters['B_end']:
        penalty_violation += (parameters['B_end'] - current_soc) * 20000

    # Global weighted Multi-Criteria Objective Z
    Z = w1 * total_time + w2 * total_cost + w3 * total_anxiety_penalty +
       penalty_violation
    return Z
```

The following Python snippet documents the chromosome fitness evaluation block coupled with dynamic exterior penalty functions used to eliminate battery floor and destination SOC payload violations.

Listing A.2: Leg Orchestration Loop inside pipeline.py

```
def execute_multileg_routing_pipeline(waypoints, vehicle_profile,
    user_preferences):
    # Orchestrates the multi-destination data flow and invokes the EA optimizer
        .

    geocoded_nodes = geocode_waypoints(waypoints)
    charging_station_db = load_epdk_database()
    full_journey_summary = []

    for leg_idx in range(len(geocoded_nodes) - 1):
        origin = geocoded_nodes[leg_idx]
        destination = geocoded_nodes[leg_idx + 1]

        # Spatial corridor pre-filtering layer
        corridor_stations = filter_corridor_bounding_box(origin, destination,
            charging_station_db)
        cost_matrix = build_routing_cost_matrix(origin, destination,
            corridor_stations, user_preferences)

        # Invoke Evolutionary Optimization Core
        optimized_leg_solution = run_ea_optimization(cost_matrix,
            user_preferences)
        full_journey_summary.append(optimized_leg_solution)

    return full_journey_summary
```

The following snippet represents the core routing loop responsible for executing geocoding steps, corridor filtering, and triggering independent leg optimizations chronologically.